



Welcome to CSC 365

Internet Programming

Dr. Yifan Zhang

1. What is HTML?

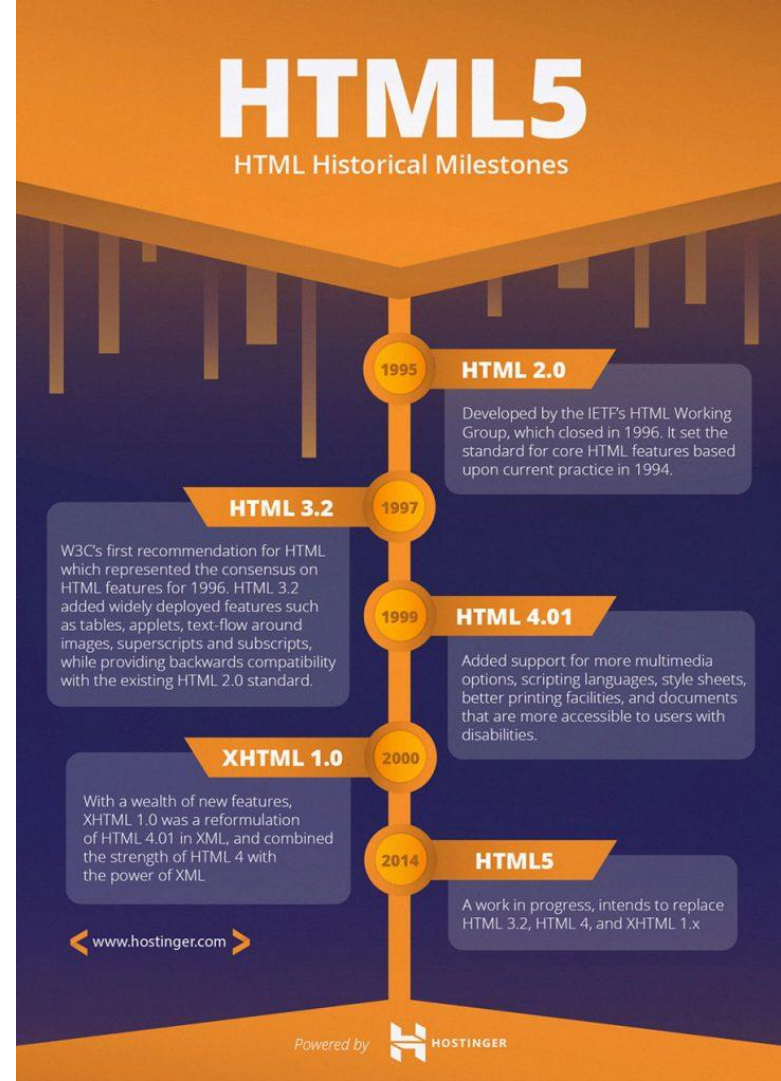
- HTML is short for Hypertext Markup Language and is the standard markup language for creating Web pages.
- HTML elements are the building blocks of HTML pages.
- HTML elements are represented by tags.
- HTML tags label pieces of content such as "heading", "paragraph", "table", and so on.
- Browsers do not display the HTML tags, but use them to render the content of the page. Content will be displayed, but tag will not be displayed.

HTML History

- First version: 1993
- HTML 3.2: Tables, text-flow around image...
- HTML 4.01: multimedia options, scripting languages...
- XHTML 1.0: XML improvements
- HTML 5: e.g. SVG, canvas and other virtual vector graphics are supported in HTML5, whereas in HTML, using vector graphics was only possible by using it in conjunction with different technologies like Flash, VML, and Silver-light, etc. (we will introduce canvas)

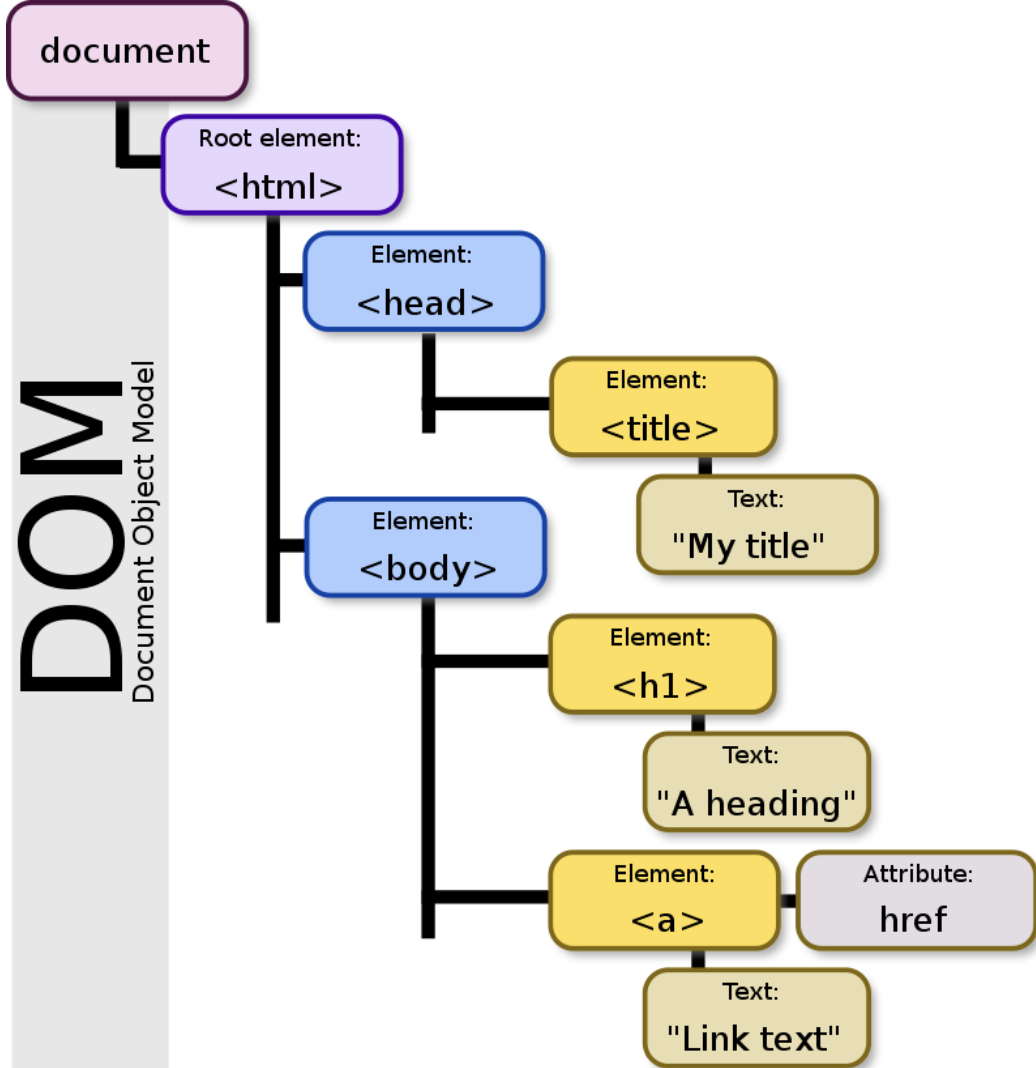
Figure Credit:

<https://www.hostinger.com/tutorials/difference-between-html-and-html5#gref>



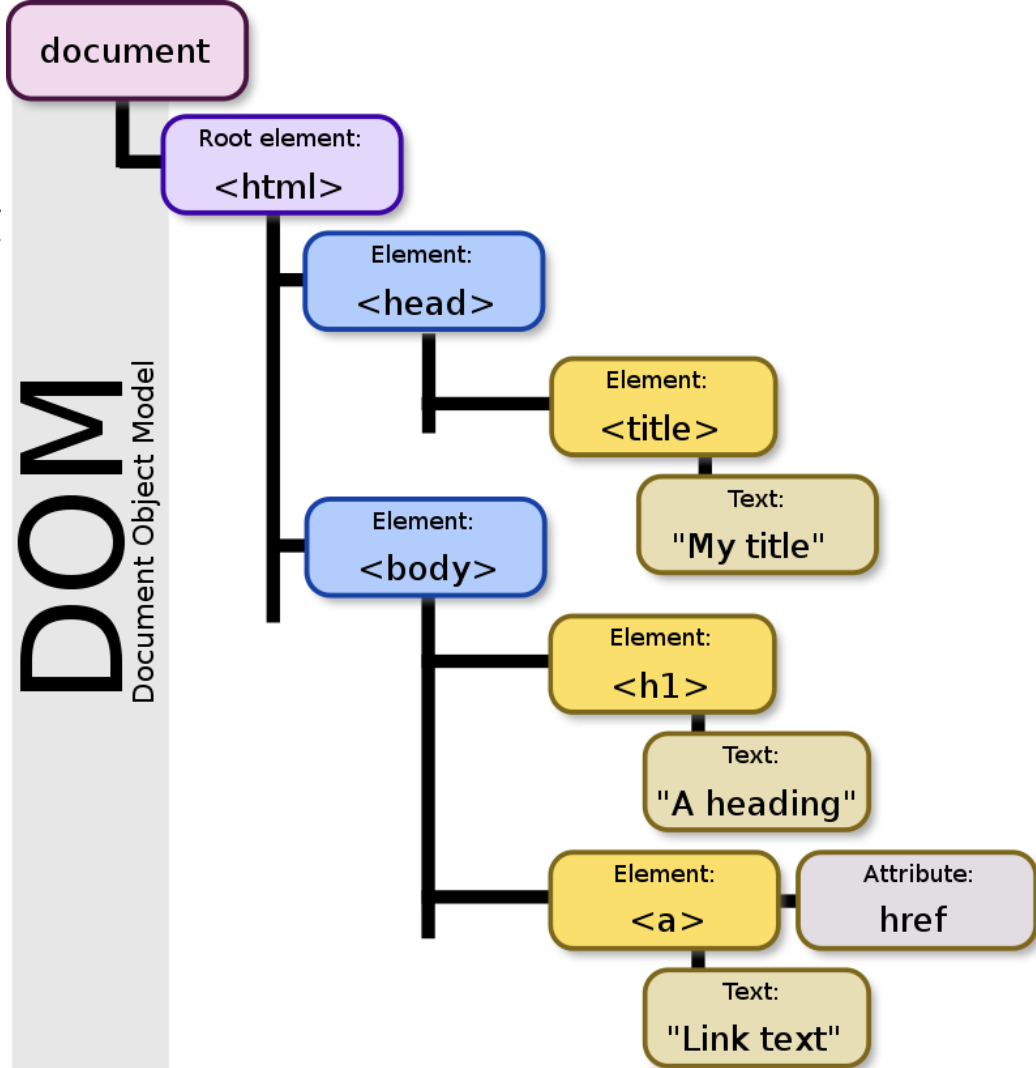
HTML

- Notice the structure?
- We call this the DocumentObject Model (DOM).
- This will be very useful later!



HTML

- Document: hierarchical collection of elements, starting with `<html>`
- Element: start tag, contents, end tag
- Elements may be nested



HTML

Tags we should know...

- **!DOCTYPE**: Defines the document type
- **html**: encapsulate entire html document
- **head**: include metadata, such as CSS and JS scripts
- **body**: main HTML content

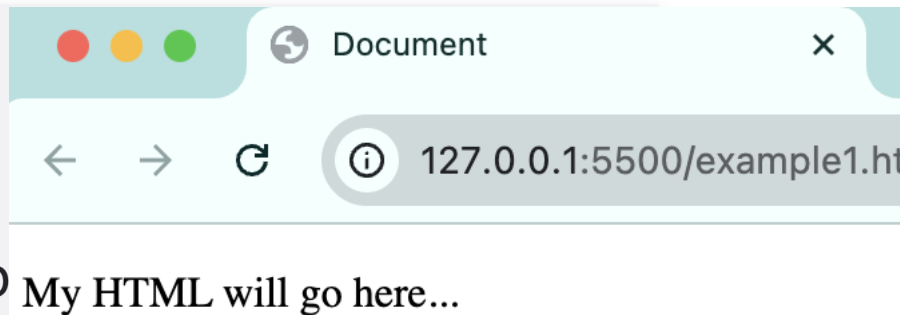
```
<html lang="en">
  <head></head>
  <body>
    <p>My HTML would go here...</p>
  </body>
</html>
```

HTML

Tags we should know...

- **!DOCTYPE**: Defines the document type
- **html**: encapsulate entire html document
- **head**: include metadata, such as CSS and JS scripts
- **body**: main HTML content

```
<html lang="en">  
  <head></head>  
  <body>  
    <p>My HTML wo  
  </body>  
</html>
```



HTML

Tags we should know...

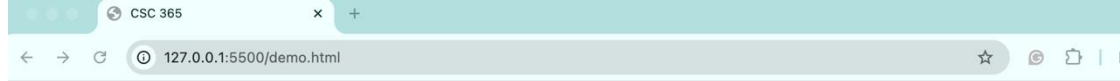
- `h1` through `h6`: headers
- `p`: paragraph text
- `strong` and `em` (instead of `b` and `i`): connotation
- `a`: anchor text (links)
- `img`: images (always include an alt attribute!)

Example 1

<> example1.html >  html

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>CSC 365</title>
7  </head>
8  <body>
9      <h1>Welcome to CSC 365</h1>
10     <p>We are going to <strong>learn skills</strong> in <em>this</em> class</p>
11     <a href="https://computerscience.missouristate.edu/">Visit Computer Science Department!</a>
12     <h2>Second head</h2>
13     <p>We will write code in this class</p>
14     
15 </body>
16 </html>
```

Example 1



Welcome to CSC 365

We are going to **learn skills** in *this class*

[Visit Computer Science Department!](https://computerscience.missouri.edu)

Second head

We will write code in this class



```
<> example1.html > html
```

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1">
6      <title>CSC 365</title>
7  </head>
8  <body>
9      <h1>Welcome to CSC 365</h1>
10     <p>We are going to <strong>learn skills</strong> in <em>this class</em></p>
11     <a href="https://computerscience.missouri.edu">Visit Computer Science Department!</a>
12     <h2>Second head</h2>
13     <p>We will write code in this class</p>
14     
15 </body>
16 </html>
```

HTML

Tags we should know...

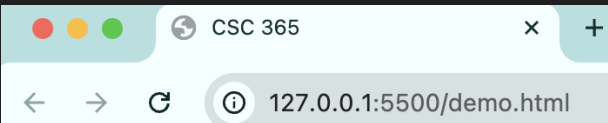
- `br`: linebreak
- `ul`, `ol`, `li`: lists
- `div`: divisions (sections)
- `!--...--` : comment

```

1  <!DOCTYPE html>
2  <html lang="en">
3
4  <head>
5      <meta charset="UTF-8">
6      <meta name="viewport" content="width=device-width, initial-scale=1.0">
7      <title>CSC 365</title>
8      <style>
9          .myDiv {
10             border: 5px outset red;
11             background-color: lightblue;
12             text-align: center;
13         }
14     </style>
15 </head>
16
17 <body>
18     <h1>Welcome to <!--CSC 365--></h1>
19     <p>The following is <br> a unordered list</p>
20     <ul>
21         <li>Springfield</li>
22         <li>Kansas City</li>
23         <li>St. Louis</li>
24     </ul>
25     <p>The following is an ordered list</p>
26     <ol>
27         <li>Joplin</li>
28         <li>Branson</li>
29     </ol>
30
31     <div class="myDiv">
32         <h2>This is a heading in a div element</h2>
33         <p>This is some text in a div element.</p>
34     </div>
35
36     <p>This is some text outside the div element.</p>
37 </body>
38
39 </html>

```

```
1 <!DOCTYPE html>
2 <html lang="en">
3
4 <head>
5   <meta charset="utf-8">
6   <meta name="viewport" content="width=device-width, initial-scale=1">
7   <title>CSC 365</title>
8   <style>
9     .myDiv {
10       border: 1px solid black;
11       background-color: #e0f0ff;
12       text-align: center;
13     }
14   </style>
15 </head>
16
17 <body>
18   <h1>Welcome to</h1>
19   <p>The following is</p>
20   <ul>
21     <li>Springfield</li>
22     <li>Kansas City</li>
23     <li>St. Louis</li>
24   </ul>
25   <p>The following is an ordered list</p>
26   <ol>
27     <li>Joplin</li>
28     <li>Branson</li>
29   </ol>
30
31   <div class="myDiv">
32     <h2>This is a heading in a div element</h2>
33     <p>This is some text in a div element.</p>
34   </div>
35   <p>This is some text outside the div element.</p>
36   <p>This is some text outside the div element.</p>
37 </body>
38
39 </html>
```



Welcome to

The following is
a unordered list

- Springfield
- Kansas City
- St. Louis

The following is an ordered list

1. Joplin
2. Branson

This is a heading in a div element

This is some text in a div element.

This is some text outside the div element.

HTML

Tags we should know...

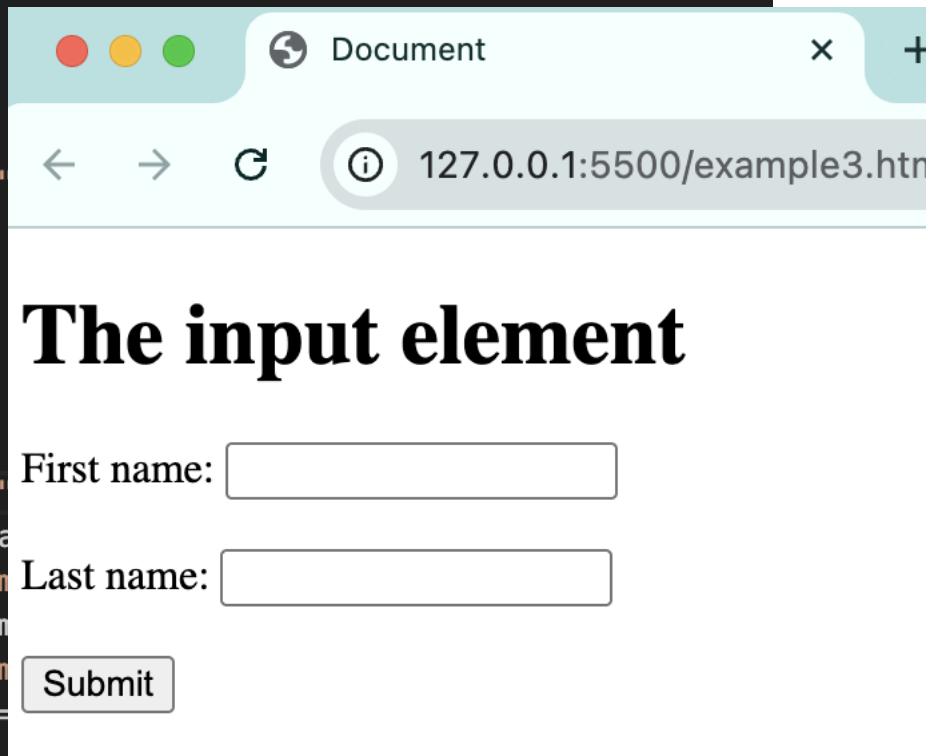
- **form**: encapsulates many inputs
- **label**: label an input (always include a for attribute)
- **button**: buttons
- **input**: user input (text, radio, button)

<> example3.html > html > body > form

```
1  <!DOCTYPE html>
2  <html lang="en">
3
4  <head>
5      <meta charset="UTF-8">
6      <meta name="viewport" content="width=device-width, initial-scale=1.0">
7      <title>Document</title>
8  </head>
9
10 <body>
11     <h1>The input element</h1>
12
13     <form action="/action_page.php">
14         <label for="fname">First name:</label>
15         <input type="text" id="fname" name="fname"><br><br>
16         <label for="lname">Last name:</label>
17         <input type="text" id="lname" name="lname"><br><br>
18         <input type="submit" value="Submit">
19     </form>
20 </body>
21
22 </html>
```

<> example3.html > html > body > form

```
1  <!DOCTYPE html>
2  <html lang="en">
3
4  <head>
5      <meta charset="UTF-8">
6      <meta name="viewport" content="width=device-width, initial-scale=1">
7      <title>Document</title>
8  </head>
9
10 <body>
11     <h1>The input element</h1>
12
13     <form action="/action_page.php">
14         <label for="fname">First name:</label>
15         <input type="text" id="fname">
16         <label for="lname">Last name:</label>
17         <input type="text" id="lname">
18         <input type="submit" value="Submit">
19     </form>
20 </body>
21
22 </html>
```



HTML

Tags for table...

- **table**: Defines a table
- **caption**: Defines a table caption
- **th**: Defines a header cell in a table
- **tr**: Defines a row in a table
- **td**: Defines a cell in a table

Table generator: https://www.tablesgenerator.com/html_tables

There is more tags

- https://www.w3schools.com/tags/ref_byfunc.asp

HTML Attributes

HTML Attributes are special properties that provide additional information about an HTML element. They modify or define the behavior, appearance, or functionality of the element.

Defined in the opening tag of an element.

```
<a href="https://computerscience.missouristate.edu/">Visit Computer Science Department!</a>
```

Written as a **name**—**value** pair:

- The name is the attribute (e.g., href).
- The value is enclosed in quotes (e.g., "https://computerscience.missouristate.edu/").

```

```

HTML Formatting Elements

Formatting elements were designed to display special types of text:

- **mark:** Mark the text
- **small:** Smaller text
- **del:** Deleted text
- **ins:** Inserted text
- **sub:** Subscript text
- **sup:** Superscript text

HTML Colors

HTML Colors are used to control the appearance of text, backgrounds, borders, and other elements in a webpage. They can be defined in multiple formats, giving developers flexibility in design:

Written as a **name**—**value** pair:

- The name is the attribute (e.g., style).
- The value is enclosed in quotes (e.g., "color:Tomato;").

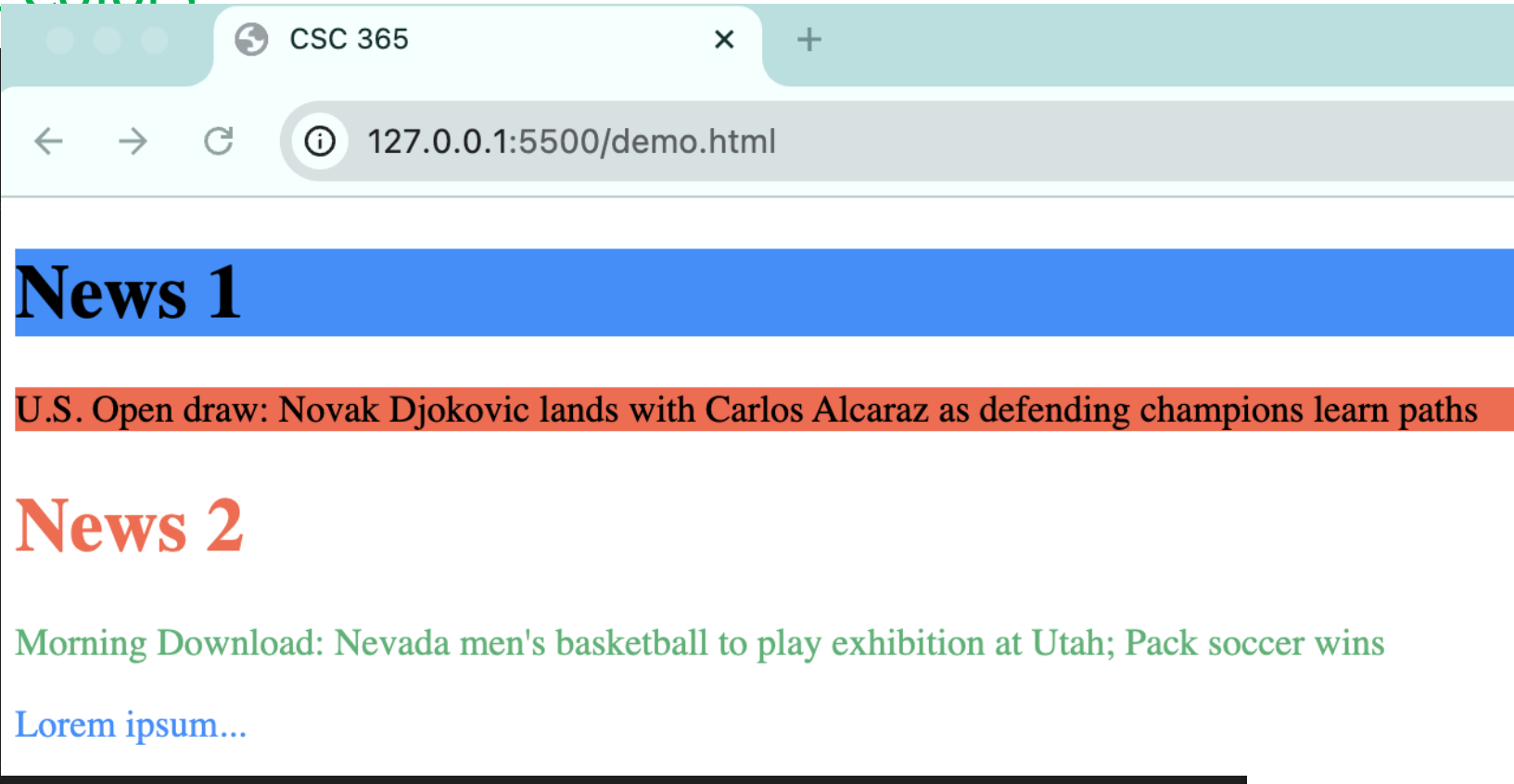
```
<h1 style="color:  Tomato;">News 2</h1>
```

HTML Colors

```
<> demo.html > html
1  <!DOCTYPE html>
2  <html lang="en">
3
4  <head>
5      <meta charset="UTF-8">
6      <meta name="viewport" content="width=device-width, initial-scale=1.0">
7      <title>CSC 365</title>
8      <style>
9          .myDiv {
10              border: 5px outset red;
11              background-color: lightblue;
12              text-align: center;
13          }
14      </style>
15  </head>
16
17  <body>
18      <h1 style="background-color: DodgerBlue;">News 1</h1>
19      <p style="background-color: Tomato;">U.S. Open draw: Novak Djokovic lands with Carlos Alcaraz as defending champions learn path
20
21      <h1 style="color: Tomato;">News 2</h1>
22      <p style="color: MediumSeaGreen;">Morning Download: Nevada men's basketball to play exhibition at Utah; Pack soccer wins</p>
23      <p style="color: DodgerBlue;">Lorem ipsum...</p>
24  </body>
25
26  </html>
```

HTML Colors

```
<> demo.html > html
1  <!DOCTYPE html>
2  <html lang="en">
3
4  <head>
5    <meta charset="utf-8">
6    <meta name="viewport" content="width=device-width, initial-scale=1">
7    <title>CSC 365
8    <style>
9      .myDiv {
10        border: 1px solid black;
11        background-color: #007bff;
12        padding: 5px;
13      }
14    </style>
15  </head>
16
17  <body>
18    <h1 style="background-color: #007bff; color: white; padding: 10px;">News 1
19    <p style="background-color: #dc3545; color: black; padding: 5px;">U.S. Open draw: Novak Djokovic lands with Carlos Alcaraz as defending champions learn paths
20
21    <h1 style="color: #dc3545; padding: 10px;">News 2
22    <p style="color: #28a745; padding: 5px;">Morning Download: Nevada men's basketball to play exhibition at Utah; Pack soccer wins
23    <p style="color: #007bff; padding: 5px;">Lorem ipsum...
24  </body>
25
26  </html>
```



In class activity 2

Try those tags! Please find in class activity 2 on BrightSpace

Next Class



Week 1:

Topic:

- HTML